

MET CS 555 – Foundation of Machine Learning

Module - 6

Logistic Regression

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Logistic Regression

Logistic Regression

In the linear regression setting, we model the relationship between **one or more explanatory variables and a continuous response variable**.

If the outcome of interest is dichotomous in nature (taking on one of two values), then **linear regression is not appropriate**.

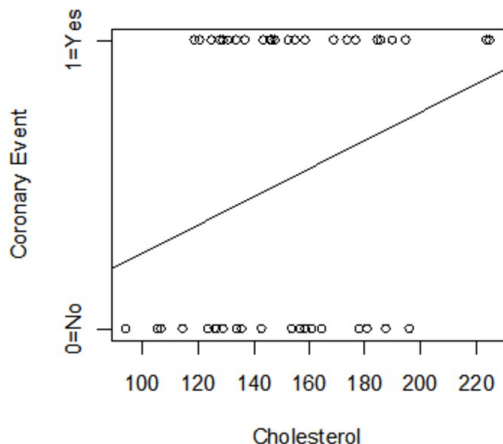
Example:

Suppose we are interested in the association between cholesterol levels and having a coronary event (a heart attack) in a high risk patient population.

In this case, our explanatory variable is cholesterol level and our outcome is whether or not the subject had another coronary event.

Logistic Regression

Many of the assumptions required for [Linear Regression](#) are not met in this setting (for example, linearity, constant variance and normality of the residuals), so inference is not valid.



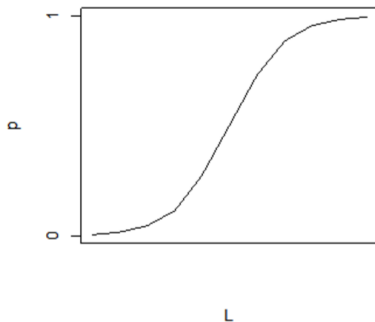
Simple Logistic Regression

The **simple logistic regression model** is based on a linear relationship between the natural logarithm (ln) of the odds of an event and a continuous explanatory variable. The equation for the simple logistic regression line is

$$L = \ln \frac{p}{1-p} = \beta_0 + \beta_1 x + E$$

where

- ▷ L is the log odds of the event
- ▷ **p is the probability of a success**
- ▷ **$\frac{p}{1-p}$ is the odds of the event**
- ▷ x is the explanatory variable
- ▷ β_0 is the intercept, β_1 is the regression coefficient, E is the random error



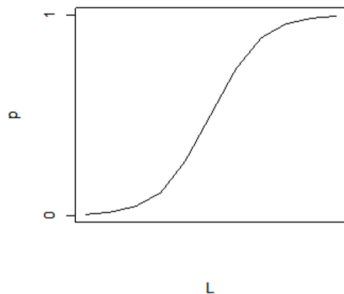
Simple Logistic Regression - Logistic / Sigmoid Function

$$L = \ln \frac{p}{1-p} = \beta_0 + \beta_1 x + E$$

If we solve the above regression equation for p , we find

$$p = \frac{e^{\beta_0 + \beta_1 x + E}}{1 + e^{\beta_0 + \beta_1 x + E}} = \frac{e^L}{1 + e^L}$$

This is called the logistic function.¹



¹https://en.wikipedia.org/wiki/Logistic_function
https://en.wikipedia.org/wiki/Sigmoid_function

Interpretation of Logistic Regression Results

Interpretation

Using the estimates of the regression coefficients, $\hat{\beta}_0$ and $\hat{\beta}_1$, **we can predict the risk of the outcome of interest for a given value of $\mathbf{x}$** using the following equation:

$$\hat{p} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x}}$$

Interpretation

$$\hat{p} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x}}$$

- ▶ The regression equation can be used to predict risk of the event.
- ▶ The interpretation of the regression coefficient(s) are generally based on odds ratios.

Consider the odds ratio of an event for a given value of $x = x_a$ versus a given value of $x = x_b$.

- ▶ The estimated odds for a given value of $x = x_a$ is given by $\widehat{\text{odds}}_a = e^{\hat{\beta}_0 + \hat{\beta}_1 x_a}$
- ▶ The estimated odds for a given value of $x = x_b$ is given by $\widehat{\text{odds}}_b = e^{\hat{\beta}_0 + \hat{\beta}_1 x_b}$

The odds ratio then is given by

$$\widehat{OR}_{x_a \text{ versus } x_b} = \frac{\widehat{\text{odds}}_a}{\widehat{\text{odds}}_b} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x_a}}{e^{\hat{\beta}_0 + \hat{\beta}_1 x_b}}$$

Interpretation

The odds of the event are $e^{\hat{\beta}_1(x_a - x_b)}$ higher for every $x_a - x_b$ unit increase in x .

Interpretation depends only on the difference in x values as opposed to their actual values.

$$\begin{aligned}\widehat{OR}_{x_a \text{ versus } x_b} &= \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x_a}}{e^{\hat{\beta}_0 + \hat{\beta}_1 x_b}} \\ &= e^{(\hat{\beta}_0 + \hat{\beta}_1 x_a) - (\hat{\beta}_0 + \hat{\beta}_1 x_b)} \\ &= e^{\hat{\beta}_0 + \hat{\beta}_1 x_a - \hat{\beta}_0 - \hat{\beta}_1 x_b} \\ &= e^{\hat{\beta}_1 x_a - \hat{\beta}_1 x_b} \\ &= e^{\hat{\beta}_1 (x_a - x_b)}\end{aligned}$$

Confidence Interval

Confidence intervals for the logistic regression setting are based on the odds ratio.

The two-sided $100\% \times (1 - \alpha)$ confidence interval for $\widehat{OR}_{x_a \text{ versus } x_b}$ is:

$$e^{\left(\hat{\beta}_1 \pm z_{\frac{\alpha}{2}} \cdot SE_{\hat{\beta}_1}\right)}(x_a - x_b)$$

Where

- ▷ $SE_{\hat{\beta}_1}$ is the standard error of the regression coefficient and
- ▷ $z_{\frac{\alpha}{2}}$ is the value from the standard normal distribution with a right tail probability of $\alpha/2$.

An Example: Logistic Regression

We are interested in the association between cholesterol levels and having a coronary event in a high-risk patient population (who have had an event in the past). We collect cholesterol data for 50 subjects and then follow each for a year to see if they have another coronary event.

Explanatory variable is cholesterol level and our outcome is whether or not the subject had another coronary event.

Given the nature of our response variable, we perform a logistic regression. A summary of the beta estimates from the model are shown below.

Parameter	Estimate	Standard Error	p-value
β_0	-3.3848	0.5838	<0.00001
β_1	0.1253	0.0362	0.0005

An Example: Logistic Regression

Parameter	Estimate	Standard Error	p-value
β_0	-3.725	1.753	0.0336
β_1	0.024	0.012	0.0420

Use these results to

- ▷ **predict the risk of another coronary event** for a high risk patient with a cholesterol level of 190.
- ▷ **calculate the odds ratio** for a coronary event of a high-risk patient with a cholesterol level of 190 versus a patient with a cholesterol level of 180.
- ▷ **calculate 95% confidence interval** for the odds ratio of having a coronary event for a patient with a cholesterol level of 190 versus a patient with a cholesterol level of 180.

An Example: Logistic Regression

Parameter	Estimate	Standard Error	p-value
β_0	-3.725	1.753	0.0336
β_1	0.024	0.012	0.0420

The risk of having a coronary event for a patient with a cholesterol level of 190 is predicted by :

$$\hat{p} = \frac{e^{-3.725+0.024*190}}{1 + e^{-3.725+0.024*190}} = \frac{e^{0.835}}{1 + e^{0.835}} = 0.697$$

An Example: Logistic Regression

The odds ratio of having a coronary event for a patient with a cholesterol level of 190 versus a patient with a cholesterol level of 180 is

$$\hat{OR}_{X_a \text{ versus } X_b} = e^{\hat{\beta}_1 (X_a - X_b)} = e^{0.024 * (190 - 180)} = e^{0.24} = 1.27$$

Interpretations:

- ▷ *The odds of having a coronary event are 1.27 times higher for every 10 Unit increase in cholesterol level.*
- ▷ *The odds ratio comparing any two individuals with cholesterol levels which are 10 units apart are the same.*

The quantity $e^{\hat{\beta}_1}$ is the odd ratio of the event for two individuals with x values that are 1 unit apart.

In other words, $e^{\hat{\beta}_1}$ is the relative increase in odds for every 1 unit increase in x.

An Example: Logistic Regression

The 95% confidence interval for the odds ratio of having a coronary event for a patient with a cholesterol level of 190 versus a patient with a cholesterol level of 180 is:

$$e^{(\hat{\beta}_1 \pm \frac{1.96}{2} \cdot SE_{\hat{\beta}_1})(x_a - x_b)} = e^{(0.024 \pm 1.96 \cdot 0.0122) \cdot 10} \\ = (1.004, 1.608)$$

We are 95% confident that the odds of having a coronary event are between 1.004 and 1.608 times higher for every 10-unit increase in cholesterol level.

Generalized linear models (GLMs)²

- ▶ We can use ANY stochastic process for generating the error, not just normal distribution.
- ▶ GLM is an extension of linear regression that allows errors to be generated by a wide variety of distributions.
- ▶ In particular, any distributions in the “Exponential Family”
- ▶ GLMs extend the linear modeling capability of R to scenarios that involve non-normal error distributions. The idea is to obtain linear functions of the predictor variables by transforming the right side of the equation by a link function.

Error Family	Link	Inverse of link	Used for
Gaussian	identity	1	normally error
Poisson	log	exp	counts
Binomial	logit	$1/(1 + 1/\exp(x))$	proportions
Gamma	inverse	$1/x$	non-constant error

²https://en.wikipedia.org/wiki/Generalized_linear_model

R Commands: Generalized Linear Models (GLMs)

Use the glm() function with binomial option

```
> glm(data$event~data$explanatory1 + data$explanatory2 + ... , family =  
      binomial)
```

- ▷ Ensure that your event is coded as 1 = *Event* and 0 = *non –event* (numeric, not a factor variable)
- ▷ If one of the variables in the model is a factor variable, it is best to create dummy variables (1/0) so that you know exactly what the reference group is
- ▷ “**family**” parameter is a simple way of specifying a choice of variance and link functions When family is set to binomial, it tells R to perform logistic regression.

<http://plantecology.syr.edu/fridley/bio793/glm.html>

R commands: Generalized Linear Models (GLMs)

```
> glm(data$event ~ data$explanatory1 + data$explanatory2 + ... , family =  
      binomial)
```

- ▷ Use the **summary()** function on the saved regression result to get regression equation and associated tests for each regression coefficient
- ▷ Use the **exp()** function, which computes the exponential value of a number e^x , on the resulting coefficients to obtain odds ratios for each regression coefficient
- ▷ Use the **predict()** function on the saved regression result to get the predicted risks for each observations

Logistic Regression: R commands

```
> data <- read.csv('cevent.csv')
# Simple logistic regression
> m <- glm(data$event ~ data$chol, family=binomial)
> summary(m)
Call:
glm(formula = data$event ~ data$chol, family = binomial)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.5752	-0.9629	-0.7217	1.1418	2.1732

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.72518	1.75307	-2.125	0.0336 *
data\$chol	0.02359	0.01160	2.034	0.0420 *

Signif.	codes:	0	***	0.001	**	0.01	*	0.05	.	0.1
		1								

...

Logistic Regression: R commands

```
# OR per 1 unit increase
> exp(cbind(OR = coef(m), confint.default(m)))
              OR          2.5 %    97.5 %
(Intercept)  0.02410882  0.000776183  0.748838
data$chol    1.02387495  1.000858995  1.047420

# OR per 10 unit increase
> exp(m$coefficients[2]*10)
data$chol
1.266103

> exp((m$coefficients[2]-qnorm(0.975)*summary(m)$coefficients[2,2])*10)
data$chol
1.008623

> exp((m$coefficients[2]+qnorm(0.975)*summary(m)$coefficients[2,2])*10)
data$chol
1.589313
```

Formal Inference in Logistic Regression

Formal Inference in Simple Logistic Regression

Formal inference in Simple Logistic Regression uses estimates of $\hat{\beta}_1$.

$H_0 : \beta_1 = 0$ (H_0 : there is no association between x and odds of the outcome)

$H_1 : \beta_1 \neq 0$ (H_1 : there is an association between x and odds of the outcome)

$\beta_1 = 0$ is equivalent to that the regression line had a slope of 0 (it would be a horizontal line), $\beta_1 = 0$ means $OR = e^{\beta_1} = 1$.

- ▷ The null hypothesis $\beta_1 = 0$ is equivalent to the test of the odds ratio for a 1 unit increase in x being equal to 1 ($H_0 : OR = 1$).
- ▷ $H_0 : \beta_1 = 0$ or $OR = 1$ is rejected if $\hat{\beta}_1$ is sufficiently far from 0.
- ▷ We reject the claim that the population parameter β_1 is equal to 0 if $\hat{\beta}_1$, the sample statistic, is far from 0.

An Example: logistic regression inference

Formally test whether or not cholesterol is associated with risk of a coronary event at the $\alpha = 0.05$ level.

1. Set up the hypotheses and select the alpha level $H_0 : \beta_1 = 0$ or $OR = 1$
(there is no association between cholesterol levels and risk for a coronary event)

$H_1 : \beta_1 \neq 0$ or $OR \neq 1$ (there is an association between cholesterol levels and risk for a coronary event) $\alpha = 0.05$

2. Select the appropriate test statistic

$$Z = \frac{\beta_1}{SE\beta_1}$$

3. State the decision rule

Determine the appropriate value from the standard normal distribution associated with a right hand tail probability of $\alpha/2 = 0.05/2 = 0.025$

$$Z_{\frac{\alpha}{2}} = 1.960$$

Decision Rule: Reject H_0 if $|z| \geq 1.96$ or Reject H_0 if $p \leq \alpha$

Otherwise, do not reject H_0

An Example: Logistic Regression Inference

4. Compute the test statistic

$$Z = \frac{\beta_1}{SE\beta_1} = \frac{0.024}{0.0116} = 2.069$$

5. Conclusion

Reject H_0 since $z \geq 1.96$ or since $p\text{-value} \leq \alpha$. We have significant evidence at the $\alpha = 0.05$ level that $\beta_1 \neq 0$. There is evidence of an association between cholesterol level and risk of a coronary event.

The odds ratio for a coronary event is $e^{\beta_1} = 1.02$ for every 1 unit increase in cholesterol. (Or we could say that the odds ratio is 1.27 for every 10-unit increase in cholesterol as this may be a more reasonable and clinically relevant scale to report the results).

We are 95% confident that the true odds ratio is between 1.00 and 1.047. (We could also report the 95% confidence interval for the 10-unit increase instead if we had chosen to present the odds ratio in the previous sentence based on this unit of increase).

R commands: Predict Method for GLM Fits

```
# predicted risk for each patient
risk <- predict(m, type=c("response"))
risk
```

1	2	3	4	5	6	7
0.22720323	0.43615030	0.34653363	0.31520904	0.23137263	0.48299515	
0.59393290						
	8	9	10	11	12	13
		14				
0.44196119	0.49478598	0.47122323	0.70593666	0.61088437	0.41309669	
0.18133869						
...						

The parameter "type" indicates the type of prediction required.

The default is on the scale of the linear predictors; the alternative "response" is on the scale of the response variable. **Thus for a default binomial model the default predictions are of log-odds (probabilities on logit scale) and type = "response" gives the predicted probabilities.**

R commands: Predict Method for GLM Fits

```
> risk <- predict(m, type=c("response"))

# predicted risk for patient with cholesterol of 190
> risk[41]
41
0.6808668

# Or manual calculation
> exp(m$coefficients[1] + m$coefficients[2] * 190) / (1 + exp(m$coefficients[1] +
  m$coefficients[2] * 190))
(Intercept)
0.6808668
```

Multiple Logistic Regression

The multiple logistic regression model is based on a linear relationship between the **natural logarithm (ln) of the odds of an event** and **a linear combination of explanatory variable**.

The equation for the full logistic regression line is given by

$$L = \ln \frac{p}{1-p} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + E$$

In the multiple logistic regression setting, **there is a global test which should be conducted first.**

An Example: Multiple Logistic Regression

Our explanatory variables are cholesterol level, age and gender and our outcome is whether or not the subject had another coronary event.

The p-value for the global test was 0.0058.

A summary of the beta estimates from the model are shown below. Test the global null hypothesis at the $\alpha = 0.05$ level.

Parameter	Estimate	Standard Error	p-value
(Intercept)	-8.536	2.684	0.001
β_{Age}	0.042	0.025	0.096
$\beta_{CholesterolLevel}$	0.029	0.013	0.024
β_{Gender}	2.521	0.803	0.002

An Example: Logistic Regression Inference

The test for the global null hypothesis tests:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1 : \text{at least one } \beta_i \neq 0$$

The p-value for the global test was 0.0058. Since , we reject the null hypothesis and conclude that there is at least one $\beta_i \neq 0$.

Example Regression Coefficient for Age:

$H_0 : \beta_{age} = 0$ or $OR_{age} = 1$ (there is no association between age and risk for a coronary event, after controlling for cholesterol level and gender)

$H_1 : \beta_{age} \neq 0$ or $OR_{age} \neq 1$ (there is an association between age and risk for a coronary event, after controlling for cholesterol level and gender)

We fail to reject the null hypothesis that or **after adjusting for cholesterol level and gender** since $p > \alpha$. We do not have significant evidence at the $\alpha = 0.05$ level that $\beta_{age} \neq 0$ ($p = 0.096$).

The odds ratio for a coronary event is 1.04 for every 1-year increase in age.

An example: logistic regression inference

Cholesterol Level:

Reject $H_0 : \beta_{chol} = 0$ or $OR_{chol} = 1$ after adjusting for age and gender since $p \leq \alpha$

We have significant evidence at the $\alpha = 0.05$ level that $\beta_{chol} \neq 0$.

There is evidence of an association between cholesterol level and risk of a coronary event after adjusting for age and gender.

The odds ratio for a coronary event is 1.029 for every 1-unit increase in cholesterol.

Gender:

Reject $H_0 : \beta_{gender} = 0$ or $OR_{gender} = 1$ after adjusting for age and cholesterol level since $p \leq \alpha$.

There is evidence of an association between gender and risk of a coronary event after adjusting for age and cholesterol level.

The odds ratio for a coronary event is 12.44 for males versus females.

An Example: Multiple Logistic Regression

Use the regression model to **predict the risk of a coronary event for a 60 year old female with a cholesterol level of 150.**

The risk of a coronary event for a 60 year old woman with a cholesterol level of 150 is 15.97%.

Note in the above equation **X_M** versus **$F = 0$** since this is the dummy variable for males (which is equal to 0 for women).

R commands: Generalized linear models (GLMs)

```
> glm(data$event~data$explanatory1 + data$explanatory2 + ... , family =  
      binomial)
```

- ▷ In multiple logistic regression, use the `wald.test()` function (from `aod` package) to get p value for the global test (of all beta coefficients = 0)

Multiple Logistic Regression

```
# multiple logistic regression
> data$male <- ifelse(data$sex == "M", 1, 0)
> m2 <- glm(data$event ~ data$chol + data$male + data$age, family =
  binomial)
> summary(m2)

# overall test
# install.package("aod")

> library(aod)
> wald.test(b=coef(m2), Sigma = vcov(m2), Terms = 2:4)

# Terms: An optional integer vector specifying which coefficients should
  be jointly tested
# Terms defines to compare which regression coefficients,
# here we want to compare the 2 to 4 (first is the intercept)
# It gives as a result Chi-Squared test results, and p-value of it
# if p is smaller than 0.05 you can reject the null hypothesis

# ORs per 1 unit increase
exp(cbind(OR = coef(m2), confint.default(m2)))
```

Model Fitness in Logistic Regression

Predictions of Logistic Regression

- ▷ In the logistic regression setting, the observed response takes on one of two values (0 = no event and 1 = event).
- ▷ The predicted probabilities take on values between 0 and 1.

If the predicted probabilities for someone who had the event are close to or equal to 1, then we would say that the model was successful in predicting that individual's risk.

How to quantify the fitness in Logistic Regression?

For example, let's say that all predicted probabilities are **0.32** or **0.72**.

- ▷ What would we say about the fit of the model if all the events had a predicted probability of **0.72** and all the non-events had a predicted probability of **0.32**?

We'd like be inclined to think that the model was successful in discriminating between those with and without the event.

- ▷ What if most of those with a predicted probability of **0.32** did not have the event and most of those with a predicted probability of **0.72** did?

We'd still be inclined to say that the model fit was good.

How we define “most” may vary and it becomes a little more difficult to assess without having a measure to quantify the fit.

Cutoffs Problem - An Example

We are interested in the association between gender and risk of having a coronary event in a high-risk patient population (who have had an event in the past).

We conduct a simple logistic regression model with gender (specifically a dummy variable for male versus female) as the only explanatory variable.

From the regression model predicted risk for a male: 64% and for a female 24%

If we have a cutoff of 0% or 100% then

- ▷ we predicted that those with a predicted probabilities of bigger than 0% will have the event and assign them predicted outcomes of **1** ($\hat{y}=1$) and
- ▷ we predicted that those with a predicted probabilities of $\leq 100\%$ will not have the event and assign them predicted outcomes of **0** ($\hat{y}=0$).

Then we could assess the fit of the predicted values versus the observed values.

The cutoff can be different.

True/False Positive and True/False Negative

- ▷ **True positive.** Rate of **Correct** identification of **Positive** category.
Example: Sick people correctly identified as sick
- ▷ **False positive.** Rate of **Incorrect** identification of **Positive** category.
Example: Healthy people incorrectly identified as sick
- ▷ **True negative.** Rate of **Correct** identification of **Negative** category.
Example: Healthy people correctly identified as healthy
- ▷ **False negative.** Rate of **Incorrect** identification of **Negative** category.
Example: Sick people incorrectly identified as healthy

In general, Positive = identified and negative = rejected.

Therefore:

- ▷ **True positive = correctly identified**
- ▷ **False positive = incorrectly identified**
- ▷ **True negative = correctly rejected**
- ▷ **False negative = incorrectly rejected**

Example

- ▷ We have identified 30 sick people **correctly identified** as sick.
- ▷ We have identified 4 healthy people **incorrectly identified** as sick.
- ▷ We have identified 25 healthy people **correctly identified** as healthy.
- ▷ We have identified 3 sick people **incorrectly identified** as healthy.

Example

- ▷ We have identified 30 sick people **correctly identified** as sick.
 - ▷ We have identified 4 healthy people **incorrectly identified** as sick.
 - ▷ We have identified 25 healthy people **correctly identified** as healthy.
 - ▷ We have identified 3 sick people **incorrectly identified** as healthy.
-
- ▷ **True Positive:** 30
 - ▷ **False Positive:** 4
 - ▷ **True Negative:** 25
 - ▷ **False Negative:** 3

Example - Confusion Matrix

- ▷ We have identified 30 sick people **correctly identified** as sick.
- ▷ We have identified 4 healthy people **incorrectly identified** as sick.
- ▷ We have identified 25 healthy people **correctly identified** as healthy.
- ▷ We have identified 3 sick people **incorrectly identified** as healthy.

- ▷ **True Positive (TP):** 30
- ▷ **False Positive (FP):** 4
- ▷ **True Negative (TN):** 25
- ▷ **False Negative (FN):** 3

		True Conditions	
		Condition Positive	Condition Negative
Predicted Class	Positive	TP=30	FP=4
	Negative	FN=3	TN=25

Sensitivity & Specificity

Sensitivity (also called Recall, Hit Rate, or true positive rate (TPR)) **measures the proportion of actual positives** that are correctly identified as such (e.g., the percentage of sick people who are correctly identified as having the condition).

$$\text{Sensitivity or Recall} = \frac{TP}{TP + FN}$$

Specificity (also called Selectivity or true negative rate (TNR)) **measures the proportion of actual negatives** that are correctly identified as such (e.g., the percentage of healthy people who are correctly identified as not having the condition).

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Precision

Precision (also called positive predictive value) is the **fraction of relevant instances among the retrieved instances**.

$$\text{Precision} = \frac{TP}{TP + FP}$$

F-Measure, F1 Measure

F-Measure provides a way to **combine Precision and Recall** as a measure of the overall effectiveness of a Classification algorithm.

F-Measure is calculated as a ratio of the weighted importance of Precision and Recall. The general formula for **positive real β number** is:

$$F_{\beta} = (1 + \beta^2) * \frac{Precision \times Recall}{(\beta \times Precision) + Recall}$$

“F Measure measures the effectiveness of retrieval with respect to a user who attaches β times as much importance to recall as precision”
(Rijsbergen et al.)

Using Precision and Recall Formula:

$$F_{\beta} = \frac{(1 + \beta^2) \times TP}{(1 + \beta^2) \times TP + \beta^2 \times FN + FP}$$

F1 Measure, F2, F0.5

F1 Score: A measure that combines precision and recall is the **harmonic mean** of precision and recall.

$$\text{F1 Score} = 2 * \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- ▷ F_2 weighs recall higher than precision (by placing **more emphasis on false negatives**)
- ▷ $F_{0.5}$ weighs recall lower than precision (by attenuating the **influence of false negatives**)

Example

- ▷ **True Positive:** 30
 - ▷ **False Positive:** 4
 - ▷ **True Negative:** 25
 - ▷ **False Negative:** 3
-
- ▷ What is the Sensitivity or Recall for this example?
 - ▷ What is the Specificity for this example?
 - ▷ What is the F1-Score for this example?

Example

- ▷ **True Positive (TP):** 30
- ▷ **False Positive (FP):** 4
- ▷ **True Negative (TN):** 25
- ▷ **False Negative (FN):** 3

$$\text{Sensitivity or Recall} = \frac{TP}{TP + FN} = \frac{30}{30 + 3} = 0.9091 \text{ or } 90.9\%$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{25}{25 + 4} = 0.8621 \text{ or } 86.21\%$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{30 + 4} = 0.8823 \text{ or } 88.23\%$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.88 \times 0.91}{0.88 + 0.91} = 0.89$$

Receiver Operating Characteristic Curve (ROC Curve)

ROC (Receiver Operating Characteristic)

Sensitivity and specificity are important measure in assessing the fit of a logistic regression model.

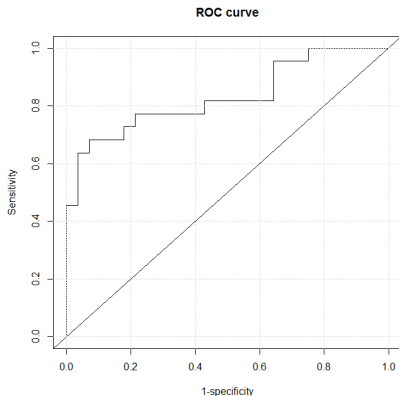
Models with one or more continuous explanatory or independent variables have more possible values for the predicted probabilities and therefore **there are often many cutoffs that produce distinct values of sensitivity and specificity.**

The area under the **ROC (receiver operating characteristic) curve (also known as the c-statistic)** is a measure of the sensitivity and specificity across the range of all possible cutoffs.

It is used to measure the goodness of fit of a logistic regression model.

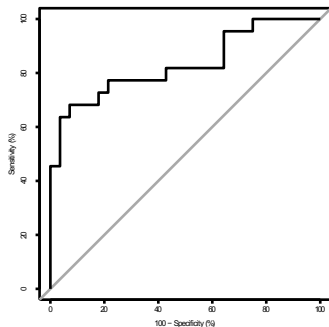
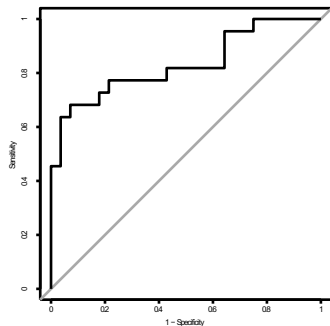
ROC (Receiver Operating Characteristic)

- ▷ The ROC curve is a plot of corresponding pairs of **sensitivity (y-axis)** and **1 minus the specificity (x-axis)** for each possible cutoff point.
- ▷ It ranges between 0.5 and 1.0 with larger values indicating better fit.
- ▷ When the area under the curve is equal to 0.50, it is said that the model **does no better at classifying events than at random or by chance.**

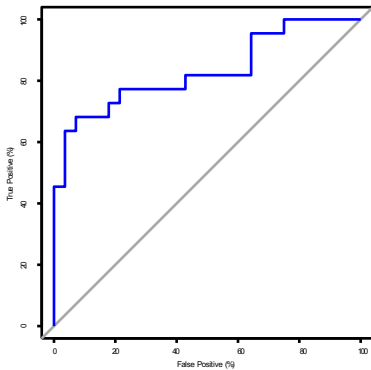
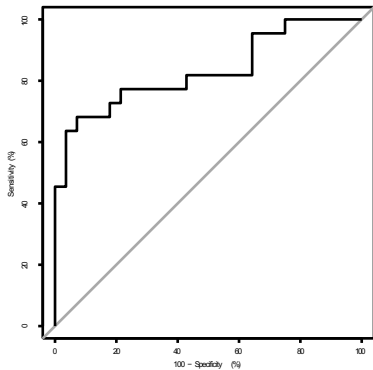


Area under the ROC curve: R commands

```
# ROC curve - Use the roc() function (from "pROC" package)
> library(pROC)
> data$prob <- predict(m, type=c("response"))
> g <- roc(data$event ~ data$prob)
> plot(g)
> plot(1-g$specificities, g$sensitivities, type="l", xlab="1-specificity",
      ylab="Sensitivity", main="ROC curve")
> abline(a=0, b=1)
```



ROC - False Positive Rate, True Positive Rate Rate



Multiple Logistic Regression and Area under ROC Curve

```
# Multiple logistic regression

> data$male <- ifelse (data$sex == "M", 1, 0)

> m2<- glm( data$event~data$chol + data$male + data$age, family=binomial)

> summary ( m2)
```

Multiple Logistic Regression and Area under ROC Curve

```
> m2<- glm(data$event~data$chol + data$male + data$age, family=binomial)

# ORs per 1 unit increase
> exp(cbind(OR = coef(m2), confint.default(m2)))

# ROC curve # install.package("pROC")
> library(pROC)

# using model with chol and sex and age
> data$prob <- predict(m2, type=c("response"))

# Build a ROC curve
> g <- roc(data$event ~ data$prob)

# see the results - c-statistics value
> print(g)

# plot the ROC Curve
> plot(g)
```